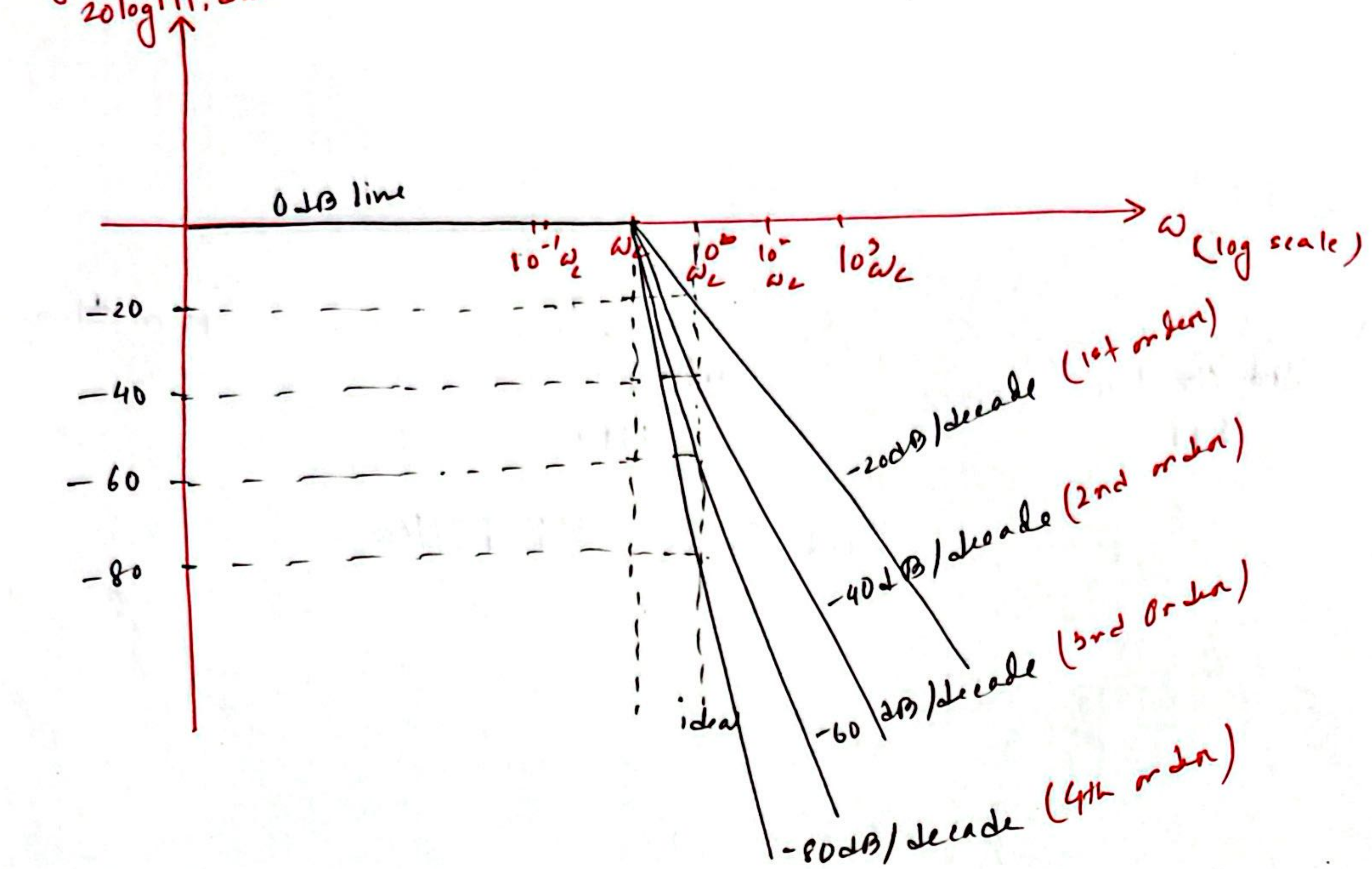
Higher Order Filters:

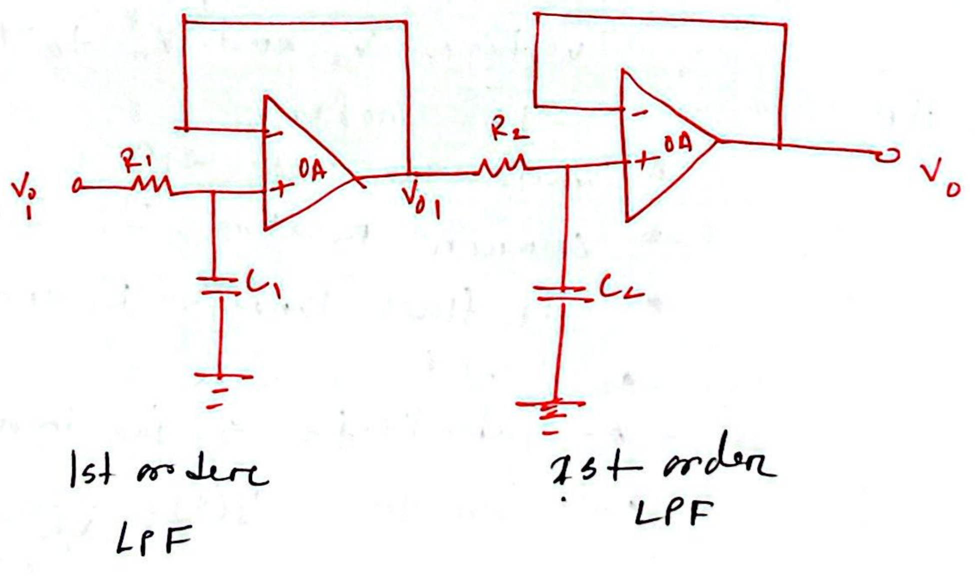


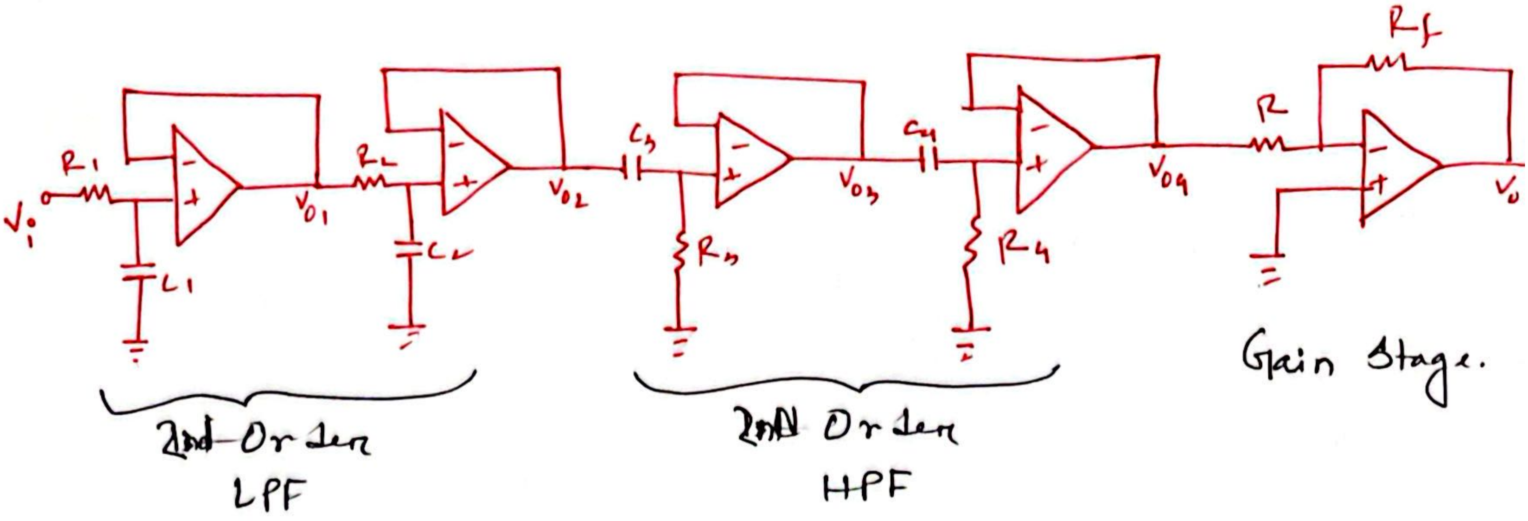
Merits:

1. Filter characteristics closer to ideal filter
2. more effective in filtering out unwanted signals, resulting in more accurate and less noisy output.

Demerits

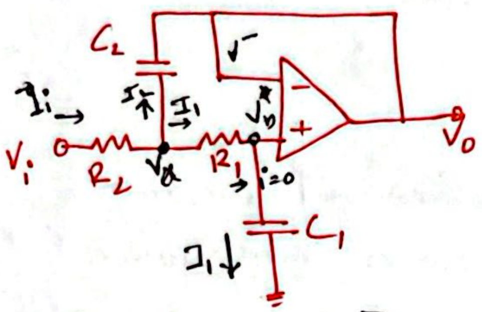
1. More circuit components
2. Consumes more power
3. Occupies more space.
4. Cost more.



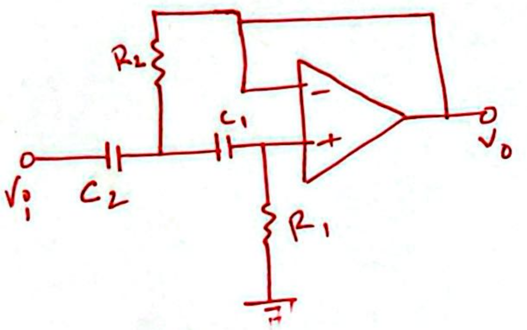


2nd Order BP Filter.

Sallen and Key filters:



2nd Order LP



2nd Order HP

Merits:

- → Use only one op-amp to construct a 2^{nd} order filter.
- → Less circuit component
- → Consumes less power
- → Consumes less space
- → Cost less.

* Identify and assign node voltages V_a and V_b^+ to the two nodes.
* Write the node eq^n
* current relation, $I_i = I_1 + I_2$
* I_1 flows through R_1 and C_1
* $V^+ = V^- = V_o$
* substitute $s = j\omega$ in node eq^n
* Solve for, $T(s) = \frac{V_o}{V_i}$

2nd Order LP filter:

$$T(s) = \frac{V_o}{V_i} = \frac{\omega_o^2}{s^2 + s\frac{\omega_o}{Q} + \omega_o^2}$$

$$\omega_o = \frac{1}{\sqrt{R_1 R_2 C_1 C_2}} \rightarrow \text{cut off freq}$$

$$\frac{\omega_o}{Q} = \frac{1}{C_2}\left(\frac{1}{R_1} + \frac{1}{R_2}\right)$$

$Q = \frac{1}{\sqrt{2}}$ For ω_o to be the cut off freq.

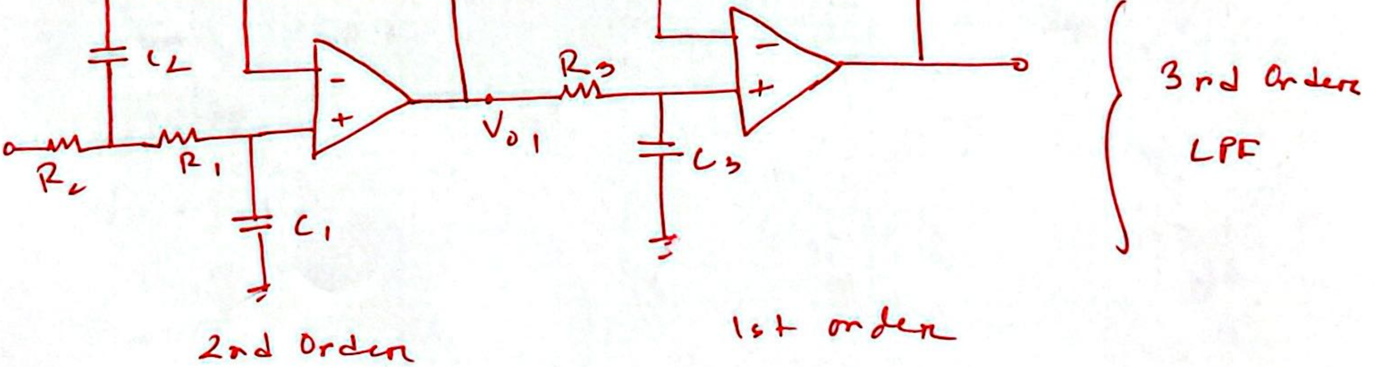
2nd Order HP filter:

$$T(s) = \frac{V_o}{V_i} = \frac{s^2}{s^2 + s\frac{\omega_o}{Q} + \omega_o^2}$$

$$\omega_o = \frac{1}{\sqrt{R_1 R_2 C_1 C_2}}$$

$$\frac{\omega_o}{Q} = \frac{1}{R_1}\left(\frac{1}{C_1} + \frac{1}{C_2}\right)$$

$$Q = \frac{1}{\sqrt{2}}.$$



3rd Order LPF $\longrightarrow$ 2nd Order LPF + 1st Order LPF

4th " " $\longrightarrow$ 2nd Order LPF + 2nd Order LPF

5th " " $\longrightarrow$ 2 2nd Order + 1 1st order LPF.

3rd Order BSF